



01

DEGREES OF FREEDOM

- For monoatomic gas, $f = 3$
- For diatomic gas,
 - (a) at room temperature, $f = 5$
 - (b) at high temperature, $f = 7$
- For triatomic gas,
 - (a) Linear $f = 5$
 - (b) Non-linear $f = 6$
- For each vibrational mode, $f = 2$

Q1

Ideal gas is composed of polyatomic molecule that has 4 vibrational modes. Total degrees of freedom is
a) 12 b) 14 c) 8 d) 6

02

SPECIFIC HEAT CAPACITY

$$\begin{aligned} \text{a) } C_p - C_v &= R \\ \text{b) } C_p - C_v &= \frac{R}{M} \quad (\text{specific heat per unit mass}) \\ \text{c) } C_v &= \frac{R}{\gamma - 1} = \frac{f}{2} R \\ \text{d) } C_p &= \frac{\gamma R}{\gamma - 1} = \left(1 + \frac{f}{2}\right) R \\ \text{e) } \gamma &= \frac{C_p}{C_v} = 1 + \frac{2}{f} \end{aligned}$$

$$\begin{aligned} \text{Mono-}\gamma &= \frac{5}{3} \\ \text{Dia-}\gamma &= \frac{7}{5} \\ \text{Tri-}\gamma &= \frac{4}{3} \end{aligned}$$

Q2

If C_p and C_v denote the specific heats of unit mass of nitrogen at constant pressure and volume respectively, then

$$\text{a) } C_p - C_v = \frac{R}{28} \quad \text{b) } C_p - C_v = \frac{R}{14} \quad \text{c) } C_p - C_v = \frac{R}{7} \quad \text{d) } C_p - C_v = R$$

03

MIXING OF GASES

$$\begin{aligned} C_{V\text{mix}} &= \frac{n_1 C_{v_1} + n_2 C_{v_2} + \dots}{n_1 + n_2 + \dots} \\ C_{P\text{mix}} &= \frac{n_1 C_{p_1} + n_2 C_{p_2} + \dots}{n_1 + n_2 + \dots} \\ \gamma_{\text{mix}} &= \frac{C_{P\text{mix}}}{C_{V\text{mix}}} \end{aligned}$$

Q3

Consider a mixture of n moles of helium gas and $2n$ moles of oxygen gas (molecules taken to be rigid) as an ideal gas. It's C_p/C_v value will be:

$$\text{a) } 19/13 \quad \text{b) } 67/45 \quad \text{c) } 40/27 \quad \text{d) } 23/15$$

04

LAW OF EQUIPARTITION OF ENERGY

$$\begin{aligned} \text{Energy for each molecule per } f &= \frac{1}{2} K_B T \\ \text{Total energy for molecule} &= \frac{f}{2} K_B T \\ \text{Monoatomic Molecule} &= \frac{3}{2} K_B T \\ \text{Total energy for a mole} &= \frac{f}{2} R T \\ \text{Total energy for } n \text{ moles} &= n f R T / 2 \\ \text{Monoatomic (1 mole)} &= \frac{3}{2} R T \\ \text{Diatomic (1 mole)} &= \frac{5}{2} R T \\ \text{Translatory Kinetic energy (1 mole, } f=3) &= \frac{3}{2} R T \end{aligned}$$

Q4

A gas mixture consists of 2 moles of O_2 and 4 moles of Ar at temperature T . Neglecting all vibrational modes, the total internal energy of the system is

$$\text{a) } 4RT \quad \text{b) } 15RT \quad \text{c) } 9RT \quad \text{d) } 11RT$$

06

VELOCITY OF GAS

$$V_{mp} : V_{avg} : V_{rms} = 1 : 1.13 : 1.225$$

07

MEAN FREE PATH

Average distance travelled by molecules between two successive collisions

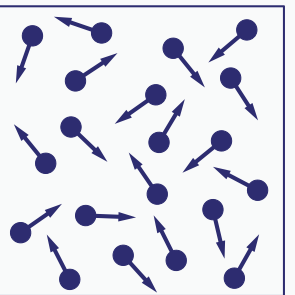
$$\begin{aligned} \lambda_{\text{mean}} &= \frac{1}{\sqrt{2} \pi d^2 n} \\ d &= \text{diameter of molecules.} \\ n &= \text{no. of molecules per unit volume} \end{aligned}$$

$$\begin{aligned} \lambda &\propto \frac{1}{d^2} \\ \lambda &\propto \frac{1}{r^2} \\ \lambda &\propto \frac{T}{P} \end{aligned}$$

05

FIRST LAW OF THERMODYNAMICS

$$\begin{aligned} Q_p &= \Delta U + W \\ \Delta U &= n C_v \Delta T \\ W &= \int P dv \\ \frac{\Delta U}{Q_p} &= \frac{1}{\gamma} \\ \frac{W}{Q_p} &= 1 - \frac{1}{\gamma} \end{aligned}$$



Root Mean square speed:

Square root of mean of square of speeds of different molecules,

$$\begin{aligned} V_{rms} &= \sqrt{\frac{V_1^2 + V_2^2 + \dots + V_n^2}{n}} \\ V_{rms} &= \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3P}{\rho}} = \sqrt{\frac{3K_B T}{m}} \end{aligned}$$

Average Speed:

Arithmetic mean of speed of molecules of gas at given temperature.

$$\begin{aligned} v_{avg} &= \frac{|\vec{V}_1| + |\vec{V}_2| + \dots + |\vec{V}_n|}{n} \\ v_{avg} &= \sqrt{\frac{8RT}{\pi M}} = \sqrt{\frac{8P}{\pi \rho}} \end{aligned}$$

Most probable speed:

Speed possessed by maximum number of molecules of gas.

$$V_{mp} = \sqrt{\frac{2RT}{M}} = \sqrt{\frac{2P}{\rho}} = \sqrt{\frac{2K_B T}{m}}$$

Q5

Consider a gas of triatomic molecules. The molecules are assumed to be triangular, made up of massless rigid rods whose vertices are occupied by atoms. The internal energy of a mole of the gas at temperature T is:

$$\text{a) } \frac{5}{2} RT \quad \text{b) } \frac{3}{2} RT \quad \text{c) } \frac{9}{2} RT \quad \text{d) } 3RT$$



Q6

The rms speeds of the molecules of Hydrogen, Oxygen & Carbon dioxide at the same temperature are V_H , V_O and V_C respectively, then:

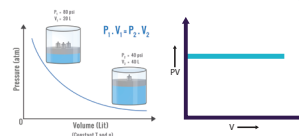
$$\begin{aligned} \text{a) } V_H > V_O > V_C & \quad \text{b) } V_C > V_O > V_H \\ \text{c) } V_H = V_O > V_C & \quad \text{d) } V_H = V_O = V_C \end{aligned}$$

Q7

The mean free path of molecules of gas, (radius r) is inversely proportional to

$$\text{a) } r^3 \quad \text{b) } r^2 \quad \text{c) } r \quad \text{d) } \sqrt{r}$$

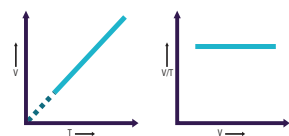
BOYLE'S LAW



$$PV = \text{constant, if } T = \text{Constant}$$

$$P_1 V_1 = P_2 V_2, \text{ when gas changes its state under constant temperature.}$$

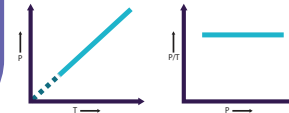
CHARLE'S LAW



$$V \propto T; \frac{V}{T} = \text{constant; } P = \text{constant.}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}, \text{ when gas change its state under constant pressure.}$$

GAY LUSSAC'S LAW



$$P \propto T; \frac{P}{T} = \text{constant; } V = \text{constant.}$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}, \text{ when gas changes its state under constant Volume.}$$

PRESSURE OF GAS

$$PV = \frac{1}{3} mn V_{rms}^2 = \frac{1}{3} mn \bar{v}^2$$

Relation between pressure and Kinetic Energy.

$$E = \frac{3}{2} PV$$

IDEAL GAS LAW

$$\begin{aligned} PV &= nRT \\ R &= 8.314 \text{ JK}^{-1} \text{mol}^{-1} \\ \rho &= \frac{PM}{RT} \end{aligned}$$

$$\begin{aligned} \text{Specific heat of Solids} &= 3R \\ \text{WATER} &= 9R \end{aligned}$$